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Excuse Me, Do You Have a Minute? An Exploration of the Dark- and Bright-Side Effects of Daily Work Interruptions for Employee Well-Being

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Work intrusions—unexpected interruptions by other people that interrupt ongoing work, bringing it to a temporary halt—are common in today's workplaces. Prior research has focused on the task-based aspect of work intrusions and largely cast intrusions as events that harm employee well-being in general and job satisfaction in particular. We suggest that apart from their task-based aspect, work intrusions also involve a social aspect—interaction with the interrupter—that can have beneficial effects for interrupted employees' well-being. Using self-regulation theory, we hypothesize that while work intrusions' self-regulatory demands of switching tasks, addressing the intrusion, and resuming the original task can deplete self-regulatory resources, interaction with the interrupter can simultaneously fulfill one's need for belongingness. Self-regulatory resource depletion and belongingness are hypothesized to mediate the negative and positive effects of work intrusions onto job satisfaction, respectively, with belongingness further buffering the negative effect of self-regulatory resource depletion on job satisfaction. Results of our 3-week experience sampling study with 111 participants supported these hypotheses at the within-individual level, even as we included stress as an alternate mediator. Overall, by extending our focus onto the social component of work intrusions, and modeling the mechanisms that transmit the dark- and the bright-side effects of work intrusions onto job satisfaction simultaneously, we provide a balanced view of this workplace phenomenon. In the process, we challenge the consensus that work intrusions harm job satisfaction by explaining why and when intrusions may also boost job satisfaction, thus extending the recent research on work intrusions' positive effects.

Keywords: work interruption, work intrusion, self-regulation, belongingness, job satisfaction

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Imagine that you are working in your office and a coworker pops in unexpectedly for a chat. You have just experienced a work intrusion—a type of work interruption that is defined as "an unexpected encounter initiated by another person that interrupts the flow and continuity of an individual's work and brings that work to a temporary halt" (Jett & George, 2003, p. 495). Given today's dynamic workplaces, work intrusions by coworkers, supervisors, and clients are very common (e.g., Leroy & Glomb, 2018; Mark, 2015; Wajcman & Rose, 2011). To date, research has largely emphasized work intrusions' negative implications for employee well-being and, in particular, for employee job satisfaction (e.g., Baethge & Rigotti, 2013; Keller et al., 2020; Perlow, 1999). However, scholars have recently challenged this negative view of work intrusions by exploring the potential positive outcomes of intrusions (e.g., Hunter et al., 2019; Sonnentag et al., 2018).

Yet, whether arguing for negative or positive outcomes of work intrusions, the emphasis of prior research has mainly been on how the interruption of an ongoing task, or the facilitation of another task, during an intrusion affects employees (e.g., Altmann & Trafton, 2002; Baethge et al., 2015; Leroy & Schmidt, 2016). This task-focused approach is certainly important, but as the opening example shows, it captures only a part of a work intrusion's core experience because work intrusions also have a social component—interaction with the interrupter. Yet, despite studies on the benefits of workplace interactions (e.g., Bhave & Lefter, 2018; Lilius, 2012), work intrusion scholars have generally lagged in recognizing that the social interaction during an intrusion may carry beneficial effects for interrupted employees (for an exception, see Ou & Davison, 2011). In a recent review, Puranik et al. (2020) noted that although scholars have generally presumed work intrusions to have negative implications for employee well-being, focusing on work intrusions' social aspect has the potential to change this conversation. To that end, our aim is to study the potential positive consequences that can arise from the social aspect of intrusions while also remaining mindful of intrusions' darker implications for employee well-being shown by past research.

To do so, we draw upon self-regulation theory, which focuses on the dynamic process wherein people respond to discrepancies between

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current and ideal states (e.g., Baumeister & Vohs, 2016). We posit that work intrusions have self-regulatory implications stemming from both their task and social aspects, resulting in two mechanisms—one for the dark side of work intrusions (self-regulatory resource depletion; Beal et al., 2005) and one for the bright side (belongingness; Leary, 2012)—that connect work intrusions to our outcome of interest: job satisfaction (widely deemed an indicator of employee well-being; Ilies et al., 2007). Our focus on job satisfaction also aligns with previous work intrusion research that has linked work intrusions to low job satisfaction (e.g., Cooper et al., 1989; Keller et al., 2020; Pachler et al., 2018).

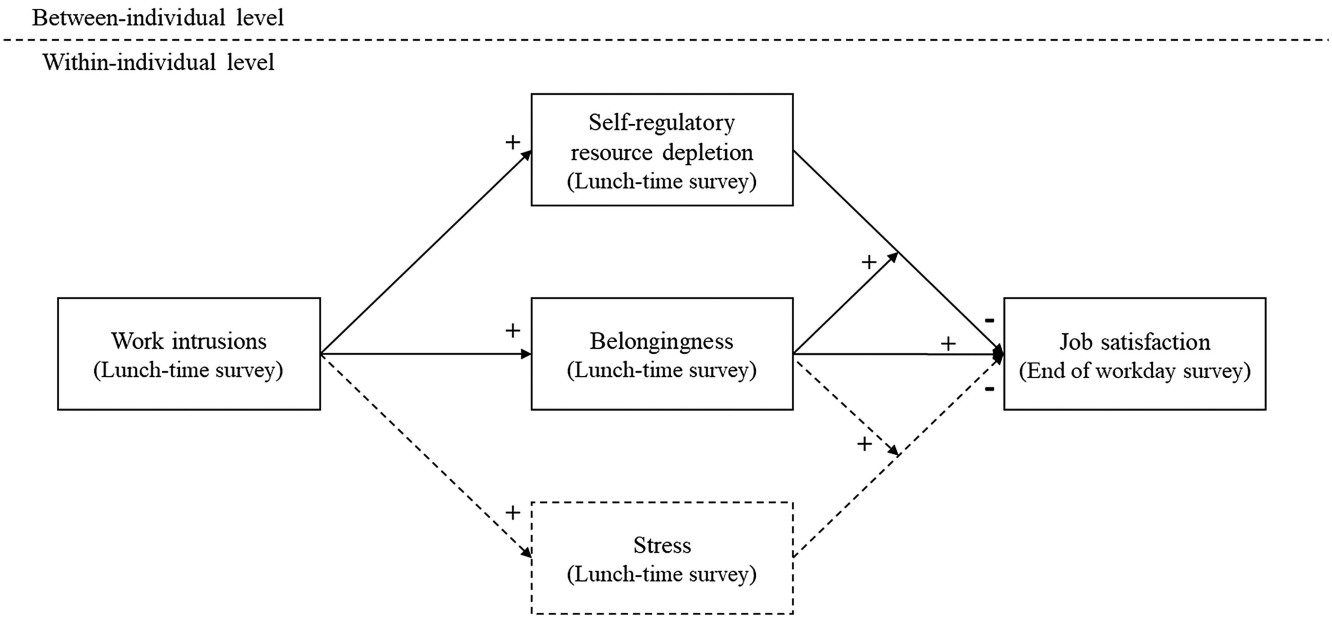
To briefly foreshadow our model, we start with its dark side. Here, the repeated shifting of attention from an ongoing task, to the cause of the intrusion, and then back to the original task during work intrusions requires self-control and draws from a limited store of self-regulatory resources (Beal et al., 2005). This may leave employees feeling depleted (Freeman & Muraven, 2010), which, in turn, should negatively affect their job satisfaction. Moving to the bright side component of our model, here we emphasize the social aspect of work intrusions. Theory on self-regulation in interpersonal contexts states that humans are social beings with an evolved need to feel as if they belong, which can be fulfilled via daily social interactions (Leary & Guadagno, 2004; Sandstrom & Dunn, 2014). Since such interactions are an integral part of work intrusions (Jett & George, 2003), we expect intrusions to have the potential to fulfill interrupted employees' need to belong, which, we posit, will positively affect their job satisfaction.

Furthermore, theory on self-regulation gives reason to believe that the above mechanisms may, in fact, interact in predicting job satisfaction. When depleted, people are less willing to expend self-regulatory resources, which can make them vulnerable to a negative state—especially in the case of work intrusions, which often hinder goals—ultimately resulting in a negative relationship between depletion and job satisfaction (Muraven et al., 2006; Wagner & Heatherton, 2013). However, theory on self-regulation (Baumeister & Vohs, 2016) and the "undoing hypothesis" (Fredrickson et al., 2000) dovetail in predicting that a positive and uplifting experience such as higher levels of belongingness can boost interrupted employees' capacity for self-regulation, thus buffering or undoing this negative relationship (Blackhart et al., 2011). Stated directly, we expect a second-stage interaction effect wherein belongingness need fulfillment mitigates the aforementioned negative relationship of self-regulatory resource depletion with job satisfaction.

We tested our theory using a within-individual approach in an experience-sampling study. As Figure 1 shows, along with the above noted relationships, we also modeled stress—a previously established mechanism of work intrusions (e.g., Jett & George, 2003)—as an alternate mediator of the link between work intrusions and job satisfaction to provide a stronger test of our hypothesized relationships. Overall, we extend the work intrusion literature in crucial ways. First, we add to our understanding of work intrusions' core experience by also emphasizing intrusions' social aspect, in addition to their task-based component. Here, we advance a new mechanism—belongingness—that not only accounts for intrusions'

Figure 1

Within-Individual, Positive, and Negative Effects of Daily Work Intrusions (i.e., Work Interruptions by Other People) on Job Satisfaction. Solid Lines Represent the Hypothesized Model, and Dotted Lines Represent the Relationships Related to the Alternate Mediating Mechanism of Stress. As Per Recommendations by ESM Scholars (e.g., Beal, 2015), We Controlled for (a) Lagged Versions of All Three Mediators and the Outcome (i.e., Their Values From the Previous Day), (b) Linear Trends (by Controlling for Study Day and Weekday), and (c) Cyclical Trends (by Controlling for the Sine and Cosine of the Weekday Variable). For Ease of Representation, We Do Not Show These Lagged and Temporal Control Variables in the Below Figure. Also, the Three Mediators Were Allowed to Covary to Capture Any Unaccounted Covariance Among Them



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social aspect but also provides a novel explanation for their positive outcomes, thus adding to the recent research that has started exploring work intrusions' positive effects (e.g., Hunter et al., 2019; Sonnentag et al., 2018).

Second, the focus on work intrusions' social aspect also helps provide, to our knowledge, the first evidence of work intrusions' positive effect on job satisfaction, thereby challenging the consensus that work intrusions tend to harm job satisfaction (e.g., Cooper et al., 1989; Keller et al., 2020; Pachler et al., 2018). This is a key insight because, in its absence, practitioners and future scholars may otherwise continue to believe that intrusions uniformly harm job satisfaction, resulting in recommendations for reducing work intrusions (e.g., Baethge & Rigotti, 2013). We show that such a conclusion may be premature, as it omits a crucial avenue to enhancing job satisfaction via increased belongingness. We thus provide a new direction for research on the relationship between work intrusions and job satisfaction and also add job satisfaction to the list of outcomes positively affected by work intrusions (Jett & George, 2003; Sonnentag et al., 2018).

Third, we extend recent research, which has modeled work intrusions' positive and negative effects simultaneously but independently (e.g., Hunter et al., 2019; Sonnentag et al., 2018), by unpacking the interaction of these effects. We show that beyond simple additive effects, the mechanisms transmitting work intrusions' impact may have a more complicated relationship with outcomes. Thus, we emphasize that a comprehensive understanding of work intrusions' impact requires the joint consideration of both the bright- and dark-side effects of work intrusions. Taken together with our elucidation of the above mediating effects, we explain *why*, and *when*, work intrusions affect job satisfaction (Whetten, 1989). In doing so, we also answer the call for an integration of interaction effects in the study of dynamic well-being (Sonnentag, 2015).

Fourth and finally, we also add to the research on self-regulation focused on uncovering factors that counteract the effect of workplace depletion (e.g., Lanaj et al., 2016; Lilius, 2012). By showing that in a workplace setting, belongingness need fulfillment can offset the effects of self-regulatory resource depletion at the daily, within-individual level, we complement previous studies that have found person-level factors that buffer the effect of daily depletion (e.g., Rosen et al., 2016). That is, we show that even in the absence of these person-level factors, managers may have other levers they can pull as an antidote to the feeling of depletion in the workplace.

Work Intrusions: A Type of Workplace Interruption

Work interruptions involve the unexpected suspension of an ongoing work task (Puranik et al., 2020). In a theory piece on interruptions, Jett and George (2003) advanced a typology of four types of interruptions—work intrusions, discrepancies, breaks, and distractions. We focus here on work intrusions, which are defined as unexpected encounters initiated by other people that temporarily halt an ongoing work task (e.g., Lin et al., 2013; Rogers & Barber, 2019).

Work intrusions occur widely in today's workplaces given the emphasis on virtual collaboration, open offices, and multiteam memberships (Mark, 2015). Wajcman and Rose (2011) noted that employees experience, on average, three intrusions per hour (up to 22 intrusions daily). Scholars have also shown work intrusions to be a common and consistent feature for employees in various occupations, including engineers, knowledge workers, manufacturing employees, healthcare workers, and information technology professionals (e.g., Andreasson et al., 2017; Baethge & Rigotti, 2015; Claessens et al., 2010; Perlow, 1999). Thus, as Leroy et al. (2020) have noted, studying the experience of work intrusions is a key research endeavor.

The Consequences of Work Intrusions for Interrupted Employees

Work intrusions are not only common but also consequential. Scholars have generally focused on how the task-related aspect of work intrusions—involving the (re)switching of tasks, addressing of the intrusion, and the performance of interrupting tasks—impacts employees by increasing task completion time, causing errors, and even leading employees to forget the original task (e.g., Altmann & Trafton, 2002; Leroy et al., 2020). Also, as intrusions increase, the time available to complete ongoing tasks reduces, which can be a stressful experience (Baethge et al., 2015). As such, studies have found stress to be a crucial consequence of work intrusions, as well as a key a mediator of the effects of intrusions onto ensuing outcomes (e.g., Fonner & Roloff, 2012; Jett & George, 2003; Mark et al., 2008; Tams et al., 2015). Beyond this, multiple studies have, over the years, shown that work intrusions can also harm employee well-being by negatively affecting job satisfaction (e.g., Baethge & Rigotti, 2013; Cooper et al., 1989; Hunter et al., 2019; Keller et al., 2020; Pachler et al., 2018), leading to a consensus that work intrusions tend to harm job satisfaction. Viewed from this perspective, the extant advocacy for reducing work intrusions seems warranted (e.g., Lin et al., 2013; Perlow, 1999; Steinhilber, 2017).

Yet, there have also been hints about potential benefits of work intrusions. For example, Jett and George (2003) speculated that intrusions can at times supply task-related information, and Speier et al. (2003) argued that intrusions can improve performance on simple tasks by forcing people to block out distractions and focus more attention on the task. Recently, scholars have shown that by facilitating other tasks, intrusions can result in positive affect, even as the impeded progress on the interrupted task results in negative affect (Hunter et al., 2019; Sonnentag et al., 2018). Thus, scholars have started focusing on the potential positive consequences of work intrusions.

Extending the Focus Onto the Social Aspect of Work Intrusions

A common thread binding the above research is that scholars have primarily focused on the effects of the task-based aspect of intrusions. Yet a work intrusion is, by definition, initiated by others (Jett & George, 2003). Thus, as noted by Puranik et al. (2020), there is a, heretofore, largely overlooked social aspect of a work intrusion—interaction with the interrupter—that is a key part of its core experience. Importantly, this social aspect may have unappreciated positive consequences and so, by not considering it, scholars may have obtained only an incomplete understanding of work intrusions. Underscoring this point, Ou and Davison (2011) showed that the social interaction during work intrusions facilitated team-level trust. Extending this line of research, we unpack the individual-level, well-being implications of work intrusions for interrupted employees by considering work intrusions' social aspect alongside the task aspect.

For this purpose, we have chosen job satisfaction as our outcome because it is (a) a well-established outcome in the work intrusion literature, (b) a key indicator of employee well-being, which is of interest to us, and (c) a vital concern for managers (Hunter et al., 2019; Ilies et al., 2007; Puranik et al., 2020). As noted before, there is a consensus in the work intrusion literature that work intrusions generally harm job satisfaction (e.g., Cooper et al., 1989; Hunter et al., 2019; Keller et al., 2020; Pachler et al., 2018). Yet, as we outline below, there are reasons to believe work intrusions might at times boost job satisfaction as well, thereby challenging this consensus. However, to be mindful of the prior findings about the negative effects of intrusions on job satisfaction, we followed recommendations by Leroy et al. (2020) and adopted a balanced approach to studying work intrusions. We did this by using a self-regulation framework to model both the bright- and dark-side effects of intrusions on job satisfaction together. This provides a complete and a nuanced picture of the relationship between work intrusions and job satisfaction.

Dual Mechanisms Linking Work Intrusions to Job Satisfaction

Drawing from self-regulation theory (e.g., Baumeister & Vohs, 2016), we identify two mechanisms—self-regulatory resource depletion and belongingness need fulfillment—that reflect the task and social aspects of work intrusions, and transmit intrusions' effects onto job satisfaction. Furthermore, as prior research has shown stress to be a mechanism that can convey work intrusions' effect onto subsequent outcomes (e.g., Fonner & Roloff, 2012; Jett & George, 2003), we included stress as an alternate mediator in our model to provide a stronger test of our self-regulatory framework. We unpack this framework below, starting with the dark side pathway.

Dark Side of Work Intrusions: Work Intrusions and Self-Regulatory Resource Depletion

Self-regulation theory focuses on the dynamic process by which people pursue goals by regulating their activities based on a comparison of their current and a desired state (Austin & Vancouver, 1996), as well as the downstream effects of this process (Baumeister & Vohs, 2016). One component of this theory focuses on the notion that the regulation of thoughts, actions, and emotions requires exerting self-control on the part of the individual, which draws from a limited pool of self-regulatory resources (Beal et al., 2005). Depletion of this pool of self-regulatory resources generally leaves people with a reduced ability to regulate subsequent thoughts, actions and emotions, as they become unwilling to expend further self-regulatory resources (Muraven et al., 2006; Rosen et al., 2016). Thus, initial activities needing self-control tend to leave employees depleted and less able to regulate later activities requiring self-control (McClean et al., in press).

We submit that work intrusions involve such activities, given their nature as unexpected events that interrupt people in the middle of their tasks (Jett & George, 2003). On facing a work intrusion, employees need to unexpectedly suspend an ongoing task, switch attention from that task onto the intrusion, cognitively understand and attend to the intrusion (maybe even perform an alternative task), and then switch attention back to the original task (Altmann & Trafton, 2002). All this time, they may need to control distracting thoughts, potentially suppress thoughts related to the task they just suspended, and regulate behavior accordingly (e.g., Leroy, 2009). Each of these activities will require self-control by the employee and will thus consume self-regulatory resources (Freeman & Muraven, 2010). Another implication is that employees may feel as if the intrusion has hindered progress on a task, which may initiate a negative reaction (Carver & Scheier, 1990). But it may be inappropriate and counternormative to express this negative reaction, as workplaces often discourage such negative displays (Gabriel et al., 2020; Sutton, 2007). So, employees may feel the need to adhere to these norms by suppressing their negative feelings, which can also be depleting (DeBono et al., 2011; Trougakos et al., 2015).

Indirect evidence supports the above arguments. For example, Lin et al. (2013) suggested that work intrusions may have self-regulatory implications, and Freeman and Muraven (2010) found that lab participants needed to exert self-control to halt an interrupted task. Analogously, organizational citizenship researchers have argued that because acts such as helping tend to occur in response to a request from a coworker (Grant & Hofmann, 2011), citizenship behavior requires a similar process of ceasing one's current task and switching attention to the nature of the request (Koopman et al., 2016). Such acts of citizenship have been found to be depleting (Gabriel et al., 2018), and we expect a similar logic to apply here. Hence, we hypothesize:

> *Hypothesis 1:* At the day level, work intrusions will be positively related to self-regulatory resource depletion for interrupted employees.

The depletion of self-regulatory resources has been linked to reduced psychological well-being (Lanaj et al., 2014; Rivkin et al., 2015). We posit that in our case, it will be negatively related to a specific indicator of well-being—job satisfaction—which reflects how satisfied one is with one's job and work context (e.g., work, coworkers, etc.), and which has been shown to vary at the daily level (Ilies & Judge, 2004; Sonnentag, 2015). As noted, much previous work intrusion research has found a negative relationship between work intrusions and job satisfaction (e.g., Baethge & Rigotti, 2013; Cooper et al., 1989; Keller et al., 2020; Pachler et al., 2018). We expect this negative relationship to be partially mediated by self-regulatory resource depletion.

Self-regulation theory suggests that depleted people enter a conservation mode where they become unwilling to expend further self-regulatory resources, which impairs their capacity to regulate subsequent activities requiring self-control (Muraven et al., 2006). This impaired capacity for self-regulation, in turn, increases people's susceptibility to negative states by weakening the barriers they would otherwise have against these states (Gailliot et al., 2006; Muraven, 2008; Wagner & Heatherton, 2013). This can be especially problematic in the context of work intrusions because, as mentioned earlier, intrusions often hinder progress toward one's goals and require acts of emotion suppression—both of which may cause and augment negative reactions (e.g., Beck et al., 2017; Carver & Scheier, 1990; Trougakos et al., 2015). Furthermore, self-regulatory resource depletion can also threaten one's ability to accomplish those hindered goals in the future (Baumeister & Vohs, 2016), as well as prevent the enactment of other important work behaviors (Gabriel et al., 2018). Taken together, depletion stemming from work intrusions can thus foster frustration and foment a negative state (Lin et al., 2013; Wagner & Heatherton,

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2013). Prior research suggests that such negative states can narrow employees' focus onto the negative aspects of their job and negatively color how they view their job and work context (Elfenbein, 2007; Forgas, 1995; Schwarz & Clore, 1983). Ultimately, the result is an unfavorable appraisal of one's job and work environment, culminating in low levels of job satisfaction (Dimotakis et al., 2011; Ilies & Judge, 2002; Judge & Ilies, 2004).

Hypothesis 2: At the day level, self-regulatory resource depletion will mediate the negative indirect relationship between work intrusions and the job satisfaction of interrupted employees.

Bright Side of Work Intrusions: Work Intrusions and Belongingness

In focusing on work intrusions' bright side, we expect the social interaction occurring during work intrusions to be an avenue for employees to fulfill their need to belong—an evolutionary desire for social inclusion (Baumeister & Leary, 1995). From a self-regulatory perspective, this need reflects a desired internal state that people typically strive for (Leary & Guadagno, 2004). Scholars hold that the reduction in the discrepancy between one's current and ideal end states in terms of fulfilling the belongingness need can be achieved via social interaction (Leary, 2012). For instance, Sandstrom and Dunn (2014) found that even some small talk with the barista when ordering a coffee led to feelings of more belongingness as opposed to just ordering a coffee. Also, scholars have shown that exposure to nonverbal cues like eye contact, smiling, or nodding during social interactions can influence belongingness (DeWall et al., 2009; Pickett et al., 2004).

Since work intrusions involve another person, they create a venue for social interaction (Jett & George, 2003) and, thus, the fulfillment of belongingness needs. For instance, day-to-day workplace interactions often have a patterned or ritualistic component and foster attachment and connection among employees (Holmes, 2005; Methot et al., in press). Work intrusions, whether work or nonwork related, likely involve a similar social component of small talk (Chong & Siino, 2006) that can signal to the interrupted employee that they are viewed favorably as an interaction partner (Sandstrom & Dunn, 2014). Moreover, norms of workplace civility generally involve the expression of regret or an apology on disturbing someone, as is likely to be the case when interrupting someone in the middle of their work (Walsh et al., 2012). Thus, irrespective of the overarching purpose of the work intrusion, interrupters may tend to apologize, indicating that they value the interrupted employee's time and, in turn, promoting a sense of belonging (Leary, 2012). Even work intrusions that are explicitly task-focused (i.e., for seeking help/information or assigning of tasks) may still be a signal that the employee's knowledge and skills are valued and acknowledged by others (Richardson & Taylor, 2012), in turn, fostering a sense of belonging.

Indirect empirical evidence supports our expectation here. For example, research in social psychology found that generally people who have more daily interactions reported higher levels of belongingness (Mehl et al., 2010; Milek et al., 2018; Sandstrom & Dunn, 2014). Similar results were found within-individual, as people reported more belongingness on days when they had more social interactions than on days with less social interactions (Bernstein et al., 2018). These results mirror recent findings in organizational scholarship. For example, Bhave and Lefter (2018) found that employees who engaged in more workplace social interactions reported higher feelings of relatedness and vitality. Taken together, we expect that as the frequency of daily work intrusions increases, employees will report higher levels of belongingness, given the increased opportunity for engaging in social interactions with others. Hence, we hypothesize:

Hypothesis 3: At the day level, work intrusions will be positively related to the level of belongingness of interrupted employees.

We further expect the sense of increased belongingness associated with work intrusions to be positively related to interrupted employees' well-being in general, and more specifically, to their job satisfaction. With regard to general well-being, theory on self-regulation in interpersonal contexts notes that fulfillment of the need to belong represents an ideal state that implies one is accepted and valued by others, which is a desirable state to be in (Leary & Guadagno, 2004). As such, the experience of belongingness is characterized by positive feelings and high self-esteem that contribute to an enhanced sense of well-being (Leary et al., 1995).

As it pertains to job satisfaction specifically, a crucial input to job satisfaction appraisals is the extent to which employees see their work environment as enabling the fulfillment of their key needs and goals (Ilies et al., 2018). That is, when employees experience work events that create desirable circumstances—such as those that fulfill their important needs—they should be more likely to report satisfaction with that job (Lynch et al., 2005; Spehar et al., 2016). By providing an avenue for social interaction, work intrusions represent one such work event that holds the potential for satisfying employees' fundamental need to belong, and should thus be associated with positive effects on employees' job attitudes such as job satisfaction (Ilies et al., 2018; Leary & Guadagno, 2004; Sandstrom & Dunn, 2014). Moreover, a sense of acceptance from one's coworkers may foster a positive appraisal of those coworkers, and such coworker appraisals are a key input into how satisfied one is with one's job (Chiaburu & Harrison, 2008). In sum, the positive experience of belongingness, representing an ideal state from a self-regulatory perspective, should lead employees to appraise their job and work environment positively, resulting in high job satisfaction (Dimotakis et al., 2011; Spehar et al., 2016). We thus expect belongingness to positively mediate the link between work intrusions and job satisfaction.

Hypothesis 4: At the day level, belongingness will mediate the positive indirect relationship between work intrusions and the job satisfaction of interrupted employees.

The Undoing of Self-Regulatory Resource Depletion by Belongingness Need Fulfillment

Until now we have positioned self-regulatory resource depletion and belongingness—stemming from the task and social aspects of intrusions—as mechanisms that independently convey the effects of work intrusions onto job satisfaction. But the task and social aspects of a work intrusion together make up its core experience (Puranik et al., 2020), implying that these mechanisms may co-occur and even interact. As such, we expect belongingness to buffer the negative relationship of depletion with job satisfaction by countering the previously described reason for this relationship—depleted employees' increased vulnerability to the negative state associated with hindered goal progress and emotion regulation requirements

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during intrusions (Beck et al., 2017; Wagner & Heatherton, 2013). Below, we unpack this notion in detail, based on the intersection of two streams of research—self-regulation and the undoing hypothesis.

From a self-regulatory perspective, belongingness need fulfillment reflects an ideal state in the context of social interactions (Leary & Guadagno, 2004). An implication of achieving such an ideal state is that people no longer need to use self-regulatory resources to strive toward this state (Baumeister & Vohs, 2007). A key tenet of self-regulation theory is that the same central store of self-regulatory resources is used for different self-control activities (Beal et al., 2005). This implies that the self-regulatory resources that interrupted employees would have otherwise used in regulating toward belongingness may now become available for other acts of self-control (Baumeister & Vohs, 2016). This should bolster their store of available self-regulatory resources and their self-regulation capacity, which would, in turn, strengthen their defenses against the aforementioned negative state associated with work intrusions (Wagner & Heatherton, 2014). Thus, one way in which higher levels of belongingness may counteract the negative effect of depletion on job satisfaction is by bolstering interrupted employees' self-regulation capacity.

Beyond this, we theorize that the positive nature of the experience of belongingness itself can also counteract the negative effects of depletion on job satisfaction. Here, we draw on the "undoing hypothesis," which states that positive experiences can potentially undo or correct the detrimental effects of negative experiences (Fredrickson et al., 2000). Although this hypothesis was first proposed in the context of affective states, it is relevant here because self-regulation and affect are inextricably intertwined (Carver & Scheier, 1990; Wagner & Heatherton, 2013) and importantly, it has been used in the prediction of daily job satisfaction (Dimotakis et al., 2011).

From the standpoint of this theory, the pleasant and uplifting experience of belongingness need fulfillment should counter some of the harmful effects of depleted employees' previously described negative state (Bono et al., 2013; Fredrickson & Branigan, 2005). Indeed, experiencing belongingness is linked with increased self-esteem and positive affect (Baumeister & Leary, 1995), which tends to broaden thought-action repertoires and promote flexible cognitive thinking (Fredrickson, 2001). This can counteract the narrowing effects of the negative state that depleted employees are vulnerable to experiencing in the case of work intrusions, which may lead them to narrowly focus on only the negative aspects of their job and their work context (Elfenbein, 2007). Instead, broadening of thought-action repertoires will mean that interrupted employees may think more broadly about their jobs (e.g., instead of focusing on only negative aspects, they may also consider positive examples of their job and working conditions) (Dimotakis et al., 2011; Fredrickson & Branigan, 2005). Thus, the positive experience of belongingness need fulfillment has the potential to weaken the negative relationship between self-regulatory resource depletion and job satisfaction (Blackhart et al., 2011).

In contrast, some work intrusions may fail to fulfill interrupted employees' need for belongingness, leaving them experiencing self-regulatory resource depletion without the added buffer of belongingness need fulfillment. Here, we expect the negative relationship between self-regulatory resource depletion and job satisfaction to be magnified (Baumeister et al., 2005). The reason is that in such cases employees are likely to perceive a large discrepancy from their ideal state of belonging, requiring them to invest significant amount of self-regulatory resources toward addressing it (Lord et al., 2010; Puranik et al., 2019). This will leave them with even fewer self-regulatory resources, and potentially an even weaker barrier against the negative state associated with goal hindrance and emotion regulation requirements during work intrusions than they would otherwise have (Gailliot et al., 2006; Wagner & Heatherton, 2013). Furthermore, lower belongingness implies a lower level of acceptance by others, which can threaten one's workplace success and is a negative and disconcerting experience (Leary, 2012). Thus, lower belongingness will only add to the aforementioned negative state that depleted employees are already vulnerable to experiencing in the case of work intrusions (Leary et al., 1995; Wagner & Heatherton, 2013). Taken together, lower levels of belongingness can exacerbate the negative relationship between self-regulatory resource depletion and job satisfaction not only by making employees more susceptible to experiencing the previously described negative state associated with work intrusions, but also by adding to this negative experience itself. Ultimately, this negative state may creep into, and magnify, employees' unfavorable appraisal of their job and work context (Forgas, 1995; Schwarz & Clore, 1983), resulting in a stronger negative relationship of self-regulatory resource depletion with job satisfaction (Baumeister et al., 2005).

Overall, we expect belongingness need fulfillment to moderate the negative effect of self-regulatory resource depletion on job satisfaction, such that this effect will be weaker at higher rather than lower levels of belongingness. Furthermore, we combine this moderation argument with our above mediation hypotheses about the effects of intrusions, to note that belongingness will moderate the negative indirect effect of intrusions on job satisfaction via self-regulatory resource depletion such that this effect will be weaker at higher rather than lower levels of belongingness:

> *Hypothesis 5a:* At the day level, belongingness will moderate the relationship between self-regulatory resource depletion and job satisfaction such that this relationship will be negative at lower levels of belongingness, whereas it will not be significant at higher levels of belongingness.

> *Hypothesis 5b:* At the day level, belongingness will moderate the indirect effect of work intrusions on job satisfaction via self-regulatory resource depletion such that this indirect effect will be negative at lower levels of belongingness, whereas it will not be significant at higher levels of belongingness.

Method

We tested our model in a daily, experience sampling methodology (ESM) study. This approach helps align our theory and methodology because not only do our focal constructs vary at the within-individual level (Beal et al., 2005; Bernstein et al., 2018; Dimotakis et al., 2011; Sonnentag et al., 2018) but given its focus on the dynamic process of goal pursuit, self-regulation theory lends itself well for this methodology (Johnson et al., 2006; Klein & Kozlowski, 2000).

Sample and Procedure

This study was approved by University of Cincinnati IRB (Title: Daily work interactions and employee outcomes; Study# 2018-1580) and is the first publication from a broader data collection. We sent an

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email inviting employees of a Midwestern U.S. university and other local organizations to participate in the study for a chance to win up to $30 worth of Starbucks gift cards (e.g., Koopman et al., in press; Lanaj et al., 2018; Tepper et al., 2018). Interested persons followed a link to a sign-in survey with a consent form, and those who met the eligibility criteria (i.e., working full time with access to email during the day) were given a description of the study and they provided data on demographic and individual-level variables. To increase participation, we used a snowball technique where participants were invited to forward the recruitment email to others who may be interested in the study (e.g., Koopman et al., 2020; Rosen et al., 2016).

During the daily portion of the study, we emailed the participants two surveys every day for 3 weeks (i.e., 15 consecutive workdays). The first (lunchtime) survey was sent around midday and captured participants' experience since arriving at work. It contained the scales for work intrusions, self-regulatory resource depletion, belongingness, and our alternate mechanism of stress, and was completed, on average, at 11:54 a.m. The second (end of workday) survey, containing the scale for job satisfaction, was sent toward the end of the workday and captured participant experience since the first survey. On average, it was completed at 4:19 p.m. and the average time elapsed between the two surveys was 4 hr and 25 min. All survey items were self-reported because our focus was on employee experiences (intrusions), internal cognitive states (self-regulatory resource depletion and belongingness), and well-being (job satisfaction)—for which "the use of same-source data is perfectly acceptable" (Gabriel et al., 2019, p. 991). While self-reports may elicit common method variance (CMV) concerns, the temporal separation of constructs, as well as our interaction results, help reduce this concern (Podsakoff et al., 2003).

Overall, 120 participants (65 recruited directly by the author team and 55 via snowball)[1] enrolled in the study. In line with prior ESM research (e.g., Gabriel et al., 2018, p. 92; Rosen et al., 2016), we dropped nine participants who did not provide data on at least 3 days of the study because at least "three data points per person are statistically needed to appropriately model within-person relationships." The final sample thus included 111 participants who provided 1147 days of data (average of 10.3 days per person), which is in line with recommendations by multilevel scholars (e.g., Gabriel et al., 2019; González-Romá & Hernández, 2017) and recent ESM studies (e.g., Rosen et al., 2019; Tang et al., 2020). Participants occupied various technical, managerial, administrative, and service positions in their organizations.[2] Their average age was 35.6 years, 77% were female, and 78% were Caucasian. Average job tenure was 4.8 years and they worked on average 8.3 hr daily and 41.5 hr per week in their workplaces.[3]

Daily Within-Individual Measures

Unless mentioned, all rating scales were from 1 = *strongly disagree* to 5 = *strongly agree*. Also, with regard to internal consistency estimate, coefficient alpha has recently been criticized for its strong assumptions about scale items being both normally distributed and tau equivalent (i.e., that items contribute equally to the scale). Scholars have noted that these assumptions are rarely met empirically, which then leads coefficient alpha to be an inconsistent estimator of reliability (e.g., Geldhof et al., 2014; McNeish, 2018). So, there have been calls to report composite reliability (omega) instead. Given the general familiarity of readers with coefficient alpha, we report it below alongside an estimate of coefficient omega (e.g., Rosen et al., in press).

Work Intrusions

As part of their lunchtime survey, participants reported the work intrusions they experienced since arriving at work on a 5-item scale developed by Parke et al. (2018). Given our focus on the frequency of work intrusions, the rating scheme for this scale was from 1 = *Never* to 5 = *Constantly*. Sample items were "I was interrupted by people seeking information from me" and, "I was interrupted by people who gave or assigned a new task to me." Average Cronbach alpha and omega values were .85 and .86, respectively.

Self-Regulatory Resource Depletion

We measured self-regulatory resource depletion in the lunchtime survey using the 5-item scale validated by Johnson et al. (2014) and used in prior depletion research (e.g., Lanaj et al., 2016). Participants rated the items based on how they felt "right now." Sample items included "I feel drained right now" and "My mental energy is running low." Average Cronbach alpha and omega were .92 and .93, respectively.

Belongingness

We adapted five items from the General Belongingness Scale of Malone et al. (2012) to measure belongingness as part of the lunchtime survey. Employees rated the extent of belongingness they experienced since arriving at work.[4] Sample items included

[1] A comparison of participants recruited directly by the author team versus those recruited via the snowball technique showed that the former had, on average, longer job tenure than the latter (5.92 years vs. 3.5 years, $p = .032$). No other differences were seen with regard to our study variables or other relevant variables.

[2] Of the final 111 participants, 71 (63%) were university employees and scored higher than nonuniversity employees on work scheduling autonomy (mean of 4.01 vs. 3.54, $p = .017$) and work interaction requirements (mean of 4.36 vs. 3.93, $p = .008$) but did not differ based on our study variables or other relevant variables. We address the implications of these differences in our supplemental analyses section.

[3] Of the final 111 participants, 45 (40.5%) worked in a private office and 64 (57.6%) worked in a shared office, with 2 participants not providing information about their workspace. A comparison revealed that participants working in private offices were older than those working in shared offices (mean of 39 years vs. 33 years, $p = .009$). Beyond this, there were no differences on any of our study variables or other relevant variables.

[4] A point to note is that our belongingness scale asked about belongingness experienced "since arriving at work," while the self-regulatory resource depletion scale measured the level of depletion "right now." Theoretically, even though we captured belongingness experienced since arriving at work (as opposed to momentary belongingness), we still expect it to be able to moderate the relationship between momentary depletion and subsequent job satisfaction as these feelings of belongingness are unlikely to dissipate completely over the short time spans of our ESM design (Beal, 2015). To test this, as part of a larger survey (Texas A&M IRB Study# 2020-0236M; Title: Study on measures of emotions, personality, and behaviors at work), we asked 181 participants from Prolific who had been at work for an average of 7.8 hr to report their level of belongingness "since arriving at work today," as well as "right now." We found these two sets of ratings to be correlated at .86. In fact, given the potential for the experience of belongingness to decay over the day, our results may actually be a more conservative test of the undoing effect of belongingness on the negative relationship between self-regulatory resource depletion and job satisfaction in our model.

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"I felt connected with others at work" and "I felt isolated from others at work" (reverse scored). Average Cronbach alpha and omega both had a value of .94.

Job Satisfaction

In line with Koopman et al. (2016), we used a 4-item scale, adapted from Brayfield and Rothe (1951), to measure job satisfaction as part of the end of the workday survey. Participants rated the items based on how they felt "right now," and sample items included "I feel fairly satisfied with my present job" and "I find real enjoyment in my work." Average Cronbach alpha and omega values were .89 and .90.

Stress (Alternate Mediator)

As noted earlier, work intrusions have been linked with workplace stress (e.g., Fonner & Roloff, 2012; Jett & George, 2003; Tams et al., 2015), and so, we included stress as an alternate mechanism for work intrusions' negative effect on job satisfaction to provide a stronger test of our hypothesized relationships. Also, for completeness we included an interaction of belongingness with stress, parallel to that of belongingness and depletion. We measured stress in the lunchtime survey with a 4-item scale given by Motowidlo et al. (1986) that has been used in prior research on workplace stress (e.g., Matta et al., 2017). Employees rated the items based on their experience since arriving at work. Sample items were "My job was extremely stressful" and "Very few stressful things happened to me at work" (reverse scored). Average Cronbach alpha and omega values were .86 and .83. An important point to note is that our results remain unchanged even when stress is dropped from the model.

Control Variables

We included several control variables recommended by ESM scholars. First, we included prior levels of endogenous variables by adding lagged versions of these variables from the previous time period (i.e., the previous day) as controls to account for autoregressive effects (Beal, 2015). We also followed recommendations to control for temporal variation in our variables, which can be an alternate explanation for observed relationships (e.g., Beal & Ghandour, 2011; Beal & Weiss, 2003). Specifically, we controlled for linear trends by including a variable ranging from 1 to 5 for the day of the week (e.g., Lim et al., 2018; Rosen et al., 2016), and another one ranging from 1 to 15 for study day (e.g., Lanaj et al., 2016; Sonnentag & Starzyk, 2015). We also controlled for weekly cyclical trends by including the sine and cosine of the above weekday variable with a period of 1 week (Beal & Ghandour, 2011; Gabriel et al., 2019). Our results, however, remain unchanged even after dropping these lagged and temporal controls. For completeness, we report the results for a model with these variables.

Analytical Strategy

Given the multilevel structure of our data (days within persons), we used multilevel path analysis in Mplus 8.4 to test our hypothesized relationships. As Table 1 shows, we first verified that there was sufficient within-individual variability to justify multilevel analysis (e.g., Gabriel et al., 2019; González-Romá & Hernández, 2017). We then proceeded with multilevel path analysis, where we simultaneously modeled all the variables from Figure 1, along with our lagged and temporal controls, at the within-individual level. The mediators were covaried with each other to account for any unmeasured covariance among them (Koopman et al., 2016).

Table 1
Percentage of Within-Individual Variance Among Daily Variables

Variable (survey)	Within-individual variance (σ^2)	Between-individual variance (τ^2)	% Within-individual variance
Work intrusions (lunchtime)	.370	.224	62
Self-regulatory resource depletion (lunchtime)	.553	.603	48
Belongingness (lunchtime)	.278	.551	34
Stress (lunchtime)	.631	.628	50
Job satisfaction (end of workday)	.218	.851	20

Note. % Within-individual variance was computed as $\sigma^2/(\tau^2 + \sigma^2)$.

Within-individual predictors were centered around each employee's mean (i.e., group mean centering) (e.g., Enders & Tofighi, 2007; Hofmann & Gavin, 1998). This helps the study of within-individual relationships by controlling for between-individual confounds (Dimotakis et al., 2013). All within-individual relationships shown in Figure 1 were modeled as random slopes, while the relationships of lagged and temporal controls were modeled as fixed effects (e.g., Lanaj et al., 2016). We included the direct effect of work intrusions on job satisfaction. To test the moderation effects, we calculated the within-individual product of belongingness and the other two mediating variables (self-regulatory resource depletion and stress) (Lanaj et al., 2018).

For indirect effects, we used a bootstrap procedure with 20,000 iterations to estimate the bias corrected confidence interval for each indirect effect. Furthermore, to confirm the hypothesized conditional indirect effect of work intrusions on job satisfaction via self-regulatory resource depletion, we checked the significance of the difference in this indirect effect at higher and lower levels of belongingness (Hayes, 2015). Finally, we calculated two types of variance explained values—overall and incremental (e.g., LaHuis et al., 2014; Rosen et al., 2016; Snijders & Bosker, 1994). The former refers to the variance explained in each of our mediators and our outcome by the overall model in Figure 1 (including the control variables). It was calculated by comparing the total and residual variance between the null and the hypothesized models, respectively. For the latter, we checked the variance explained by each specific path in our model by comparing the residual variance of a model with and without that particular path.

Results

Table 2 shows the means, standard deviations, and correlations among the variables. Before testing the hypotheses, we ran a multilevel confirmatory factor analysis of the five focal variables from Figure 1 (work intrusions, self-regulatory resource depletion, belongingness, job satisfaction, and stress). This 5-factor model with

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Table 2
Means, Standard Deviations, and Within-Individual Correlations

Variable (survey)	Mean	*SD*	1	2	3	4
1. Work intrusions (lunchtime)	1.82	.61				
2. Self-regulatory resource depletion (lunchtime)	2.25	.74	.13***			
3. Belongingness (lunchtime)	4.01	.53	.08**	−.15***		
4. Stress (lunchtime)	2.40	.79	.32***	.20***	−.17***	
5. Job satisfaction (end of workday)	3.59	.47	−.07*	−.16***	.14***	−.14***

Note. Level 1 n = 1,147; level 2 n = 111. Correlations represent the group-mean centered relationships among the daily variables.
$^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

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all 23 items loading onto their corresponding factors fit the data adequately (χ^2 = 855.94, df = 220, CFI = .91, RMSEA = .05, $SRMR_{within}$ = .04). Moreover, the fit was better than a 4-factor model in which depletion and stress items were allowed to load onto a single factor (χ^2 = 2186.53, df = 224, CFI = .74, RMSEA = .09, $SRMR_{within}$ = .10; Satorra–Bentler χ^2_{diff} = 531.4, df = 4, $p < .001$), and another 4-factor model in which belongingness and job satisfaction items loaded onto a single factor (χ^2 = 1565.97, df = 224, CFI = .82, RMSEA = .07, $SRMR_{within}$ = .08; Satorra–Bentler χ^2_{diff} = 335.56, df = 4, $p < .001$), as well as a 1-factor model where all items loaded onto a single factor (χ^2 = 5744.23, df = 230, CFI = .26, RMSEA = .14, $SRMR_{within}$ = .17; Satorra–Bentler χ^2_{diff} = 1280.54, df = 10, $p < .001$). Overall, this provides support for the factor structure of our 5-factor model from Figure 1.

Tests of Hypotheses

We then proceeded with multilevel path analysis (see Table 3 for results). Starting with the dark-side pathway, we found work intrusions to be positively related self-regulatory resource depletion (γ = .150, SE = .046, p = .001), thus supporting Hypothesis 1. This indicates that the unexpected interruption and switching of tasks during work intrusions may take a self-regulatory toll on employees. Furthermore, in support of Hypothesis 2, self-regulatory resource depletion was negatively related to job satisfaction (γ = −.071, SE = .023, p = .002), and mediated the negative indirect effect of intrusions on job satisfaction (indirect effect = −.011, 95% CI [−.023, −.003]). This result reaffirms the negative relationship between intrusions and job satisfaction from prior research, while advancing self-regulatory resource depletion as a mediator of this relationship.

For the bright-side pathway, in support of Hypothesis 3, intrusions were positively related to belongingness (γ = .088, SE = .037, p = .018), meaning being interrupted by others may be an avenue to fulfill one's need to belong. Furthermore, belongingness was positively related to job satisfaction (γ = .099, SE = .032, p = .002), and mediated the indirect relationship of intrusions with job satisfaction (indirect effect = .009, 95% CI [.002, .021]), supporting Hypothesis 4. To our knowledge, this is the first evidence of a positive relationship between work intrusions and job satisfaction, implying this relationship may be more nuanced than previously thought.

For Hypothesis 5a, we found support for the interaction effect of belongingness on the relationship of self-regulatory resource depletion with job satisfaction (γ = .115, SE = .041, p = .006), such that this relationship was negative at lower levels of belongingness (γ = −.126, SE = .032, $p < .001$), but was not significant at higher levels of belongingness (γ = − .016, SE = .030, p = .591) (see Figure 2). For the conditional indirect effect, as Table 3 shows, the indirect effect of intrusions on job satisfaction via self-regulatory resource depletion was negative at lower levels of belongingness (indirect effect = −.019, 95% CI [−.038, −.006]), but was not significant at higher levels of belongingness (indirect effect = −.002, 95% CI [−.015, .005]). Also, the difference between these two indirect effect values was significant (indirect effect difference = .017, 95% CI [.004, .038]), supporting Hypothesis 5b (Hayes, 2015). Overall, our results show that belongingness not only mediated the positive effect of work intrusions on job satisfaction, but also buffered the negative effect of self-regulatory resource depletion on job satisfaction.

Importantly, our hypotheses were supported after adding stress as an alternate mediator of the relationship between intrusions and job satisfaction, indicating that our two hypothesized mediators—depletion and belongingness—have predictive capacity beyond this previously established mechanism of work intrusions. For results related to stress, as seen in Table 3, work intrusions were positively related to stress (γ = .453, SE = .056, $p < .001$), stress was negatively related to job satisfaction (γ = −.050, SE = .024, p = .039) and also mediated the negative indirect effect of work intrusions on job satisfaction (γ = −.023, 95% CI [−.046, −.002]). The moderation effect of belongingness on the link between stress and job satisfaction was not significant (γ = −.070, SE = .050, p = .166)—we discuss this in more detail in our discussion.

Finally, regarding the overall variance explained, our model from Figure 1, along with the control variables, accounted for 14.5% variance in self-regulatory resource depletion, 18% in belongingness, and 15.1% in job satisfaction. With regard to incremental variance explained by our hypothesized relationships, work intrusions explained incremental variance of 5.2% in self-regulatory resource depletion and 8.1% in belongingness, and the direct effect of self-regulatory resource depletion and belongingness each explained incremental variance of 2.1% in job satisfaction.

Supplemental Analyses

We ran several supplemental analyses to check the robustness of our findings. First, as noted in footnote 3, we found that participants working in private offices tended to be older than those working in shared offices. Apart from this difference, these two groups did not differ based on our study variables or other relevant variables (e.g., job tenure, daily and weekly work hours, task interdependence, workplace polychronicity, job complexity, work interaction requirements, and work scheduling autonomy). Furthermore, our

Table 3
Results of Multilevel Path Analysis for the Model in Figure 1

Variable	Self-regulatory resource depletion (lunchtime)		Belongingness (lunchtime)		Stress (lunchtime)		Job satisfaction (end of workday)	
	γ	*SE*	γ	*SE*	γ	*SE*	γ	*SE*
Intercept	2.25***	.078	4.01***	.072	2.40***	.079	3.59***	.091
Work intrusions (lunchtime)	.150**	.046	.088*	.037	.453***	.056	−.040	.027
Self-regulatory resource depletion (lunchtime)	—	—	—	—	—	—	−.071**	.023
Belongingness (lunchtime)	—	—	—	—	—	—	.099**	.032
Stress (lunchtime)	—	—	—	—	—	—	−.050*	.024
Self-regulatory resource depletion × belongingness	—	—	—	—	—	—	.115**	.041
Stress × belongingness	—	—	—	—	—	—	−.070	.050
Lagged controls (prior day values)								
Prior self-regulatory resource depletion	−.001	.041	—	—	—	—	—	—
Prior belongingness	—	—	.087	.048	—	—	—	—
Prior stress	—	—	—	—	.161***	.042	—	—
Prior job satisfaction	—	—	—	—	—	—	.150**	.057
Temporal controls								
Study day	.008	.006	−.002	.005	−.001	.007	−.006	.005
Weekday	−.044	.080	.038	.052	.005	.084	−.112*	.056
Sine (weekday)	−.059	.125	.034	.084	.096	.128	−.134	.087
Cosine (weekday)	−.010	.077	.029	.047	−.070	.081	−.107	.059
Indirect effects								
Through self-regulatory resource depletion								
Indirect effect	—	—	—	—	—	—	−.011* [−.023, −.003]	
Indirect effect (at high belongingness)	—	—	—	—	—	—	−.002 [−.015, .005]	
Indirect effect (at low belongingness)	—	—	—	—	—	—	−.019* [−.038, −.006]	
Indirect effect difference	—	—	—	—	—	—	.017* [.004, .038]	
Through belongingness								
Indirect effect	—	—	—	—	—	—	.009* [.002, .021]	

Note. Level 1 $n = 1{,}147$; level 2 $n = 111$; γ values reflect unstandardized coefficients. For the indirect effects, values in the brackets reflect 95% confidence interval.
* $p < .05$. ** $p < .01$. *** $p < .001$.

results were unchanged by the inclusion of this distinction as a cross-level moderator of the relationship between intrusions and our meditators.

Second, we modeled two variables that can influence norms related to work intrusions—workplace polychronicity and task interdependence—as cross-level moderators of the link between intrusions and our mediators in two separate analyses. Workplace polychronicity is the extent to which organizations stress the simultaneous performance of multiple tasks (Mattarelli et al., 2015). Its cross-level effect on the relationship between intrusions and our mediators was not significant, nor did our hypothesized results change on adding it. Research has also shown that task interdependence can influence intrusions (Perlow, 1999; Van der Vegt et al., 2001), and on adding it as a cross-level moderator of the relationship between intrusions and our mediators, our hypothesized relationships did not change. But it did moderate the relationship between work intrusions and belongingness ($\gamma = .094$, $SE = .047$, $p = .043$), such that this positive relationship was stronger when task interdependence was higher ($\gamma = .160$, $SE = .059$, $p = .007$) than when it was lower ($\gamma = .002$, $SE = .046$, $p = .973$)—a point to which we return in our discussion section.

Third, we probed the distinction between university and nonuniversity employees in terms of control over intrusions. The average number of intrusions did not vary across these two groups of employees. Plus, these groups also did not differ based on our study variables, or other relevant variables (e.g., age, job tenure, daily and weekly work hours, workplace polychronicity, task interdependence, job complexity). But as noted in footnote 2, university employees reported higher work scheduling autonomy and greater work interaction requirements than nonuniversity employees (Morgeson & Humphrey, 2006), which could be relevant to their level of control over work intrusions. We thus probed the impact of these variables by modeling them as cross-level moderators of the relationship between intrusions and our mediators in two separate analyses. Our hypothesized relationships remained unchanged in both cases. While work interaction requirements did not have an effect on the relationship between intrusions and our mediators, work scheduling autonomy did moderate the relationship

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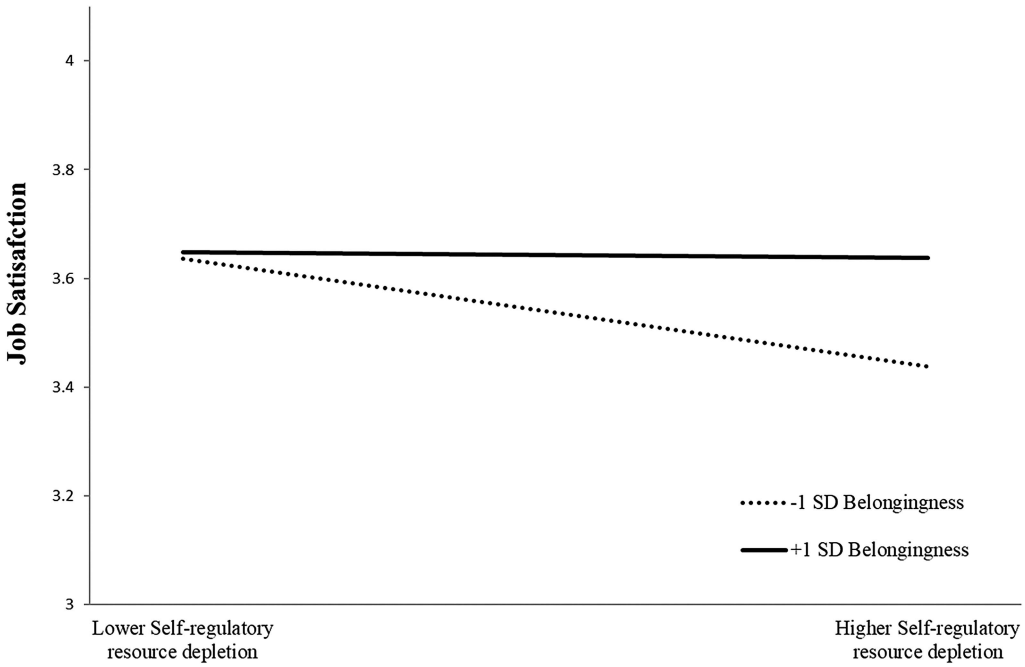
Figure 2

Moderating Effect of Belongingness on the Within-Individual Relationship Between Self-Regulatory Resource Depletion and Job Satisfaction. The x-Axis (Self-Regulatory Resource Depletion) Reflects a Range of 2 Standard Deviation (i.e., ±1 SD From the Mean for Higher and Lower Levels of Self-Regulatory Resource Depletion). The y-Axis (Job Satisfaction) Similarly Reflects a Range of Approximately 2 SDs. The Interaction Pattern Corresponds to the Predictions of Self-Regulation Theory. That Is, the Relationship Between Self-Regulatory Resource Depletion and Job Satisfaction Is Negative and Significant at Lower Levels of Belongingness (γ = −.126, SE *= .032,* p *< .001), and Not Significant at Higher Levels of Belongingness (γ = −.016, S.E. = .030, p = .591)*



between intrusions and belongingness (γ = .076, *SE* = .038, *p* = .042) such that this positive relationship was stronger when work scheduling autonomy was higher (γ = .161, *SE* = .049, *p* = .001) than when it was lower (γ = .008, *SE* = .056, *p* = .893). We discuss the implications of this finding in our discussion section.

Lastly, based on an anonymous reviewer's suggestion, we split the Parke et al. (2018) work intrusion scale into two factors[5]—factor 1 had three items for intrusions where others seek information/help or assign new tasks (e.g., "I was interrupted by people seeking information from me," "I was interrupted by people seeking my help," "I was interrupted by people who gave or assigned a new task to me"), and factor 2 had two items for nonwork intrusions or intrusions that lead to information gain (e.g., "I was interrupted by people for non-work related matters" (e.g., socializing), and "I was interrupted by people who provided me work-related updates or information")—to probe whether these factors related differently with our mediators.

First, we ran a multilevel CFA to compare our hypothesized 5-factor model (in which all intrusion items loaded onto a single factor) with a 6-factor model in which the intrusion items were allowed to load onto two different factors, as described above. While the 6-factor model with two work intrusion factors had an acceptable fit (χ^2 = 846.79, *df* = 215, CFI = .91, RMSEA = .05, $SRMR_{within}$ = .04), it was not significantly better than the more parsimonious 5-factor model with a single work intrusion factor in it (Satorra–Bentler χ^2_{diff} = 9.15, *df* = 5, *p* = .103). We did, however, run an exploratory analysis where we included these two intrusion factors as separate predictors of the mediators from Figure 1. Results showed that the two factors related differentially with our mediators—factor 1 was significantly related to depletion (γ = .105, *SE* = .049, *p* = .034), but not to belongingness (γ = .024, *SE* = .028, *p* = .390), while factor 2 was significantly related to belongingness (γ = .077, *SE* = .038, *p* = .04) but not to depletion (γ = .033, *SE* = .055, *p* = .55). All our hypothesized second-stage relationships remained unchanged.

This finding is interesting, and was something we had not considered, since prior work with this measure has also treated it as unidimensional (Parke et al., 2018). As it pertains to the theory we build and test here, we retained the more parsimonious 5-factor model (with a single factor for the work intrusion scale) because this conceptualization is consistent with prior theory and empirics in the

[5] An anonymous reviewer suggested that these two factors could be qualitatively different given the different types of work intrusions that make up these factors and can hence relate differently with our mediators. Accordingly, we ran a supplemental analysis where we added these two factor as two different predictors of our mediators from Figure 1.

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work intrusion literature (e.g., Fletcher et al., 2018; Jett & George, 2003; Lin et al., 2013; Parke et al., 2018; Rogers & Barber, 2019). Furthermore, it also reflects our theoretical focus at the outset of this study. However, the results of our exploratory analysis provide what we feel are crucial directions for future research, as we explain in more detail in the discussion.

Discussion

Being interrupted by others is a common occurrence in today's workplaces given the emphasis on open workspaces, virtual collaboration, and multitasking. Using a self-regulation lens, we showed that while the interruption and switching of tasks during intrusions can take a toll on employees' self-regulatory resources, the social interaction with the interrupter can help boost their level of belongingness. These two mechanisms were linked negatively and positively to job satisfaction, respectively. Overall, we advance a balanced understanding of intrusions that accounts for both their task and social aspects, as well as their dark- and bright-side effects.

Theoretical Contributions

Our central theoretical contribution is to the literature on work intrusions. By unpacking the consequences of the social interaction occurring during work intrusions, we provide an expanded and more complete view of the core experience of work intrusions. Past research, whether on negative or positive outcomes of intrusions, has largely ignored this social aspect of work intrusions, focusing instead on the consequences of intrusions' task-based aspect (Jett & George, 2003; Leroy et al., 2020). Our results suggest that this may have left out an influential part of work intrusions' core experience, resulting in an incomplete understanding of this workplace phenomenon. By demonstrating that work intrusions can be an avenue to fulfill one's need to belong, we highlight the importance of work intrusions' social aspect, and also advance a novel mechanism—belongingness—that explains *why* work intrusions can have positive effects.

Our second theoretical contribution lies in demonstrating a positive relationship between work intrusions and job satisfaction that is counter to most of prior work intrusion research. To our knowledge, we are the first to theorize about, and empirically show, this positive relationship based on our focus on work intrusions' social aspect. We believe this is a crucial insight because in its absence, it is possible that practitioners and future researchers may continue to believe that work intrusions uniformly harm job satisfaction, based on the vast amount of past research that has found a negative relationship between these constructs (e.g., Baethge & Rigotti, 2013; Cooper et al., 1989; Hunter et al., 2019; Keller et al., 2020; Pachler et al., 2018). Such a conclusion could, in turn, result in recommendations and interventions to reduce work intrusions, given that job satisfaction is a key concern for managers in the workplace. Indeed, we do see writings in the popular media that cast work intrusions as unwanted events and recommend elimination of work intrusions (e.g., Schulte, 2015; Steinhilber, 2017). Our findings suggest that such a conclusion may be premature, and we bring a new perspective to this relationship by showing that work intrusions can at times even enhance job satisfaction. In the process, we also extend the recent research on work intrusions' positive effects by identifying job satisfaction as a potential positive outcome of work intrusions (e.g., Hunter et al., 2019; Sonnentag et al., 2018).

Third, recent studies have provided crucial insights by modeling both the positive and the negative effects of work intrusions as parallel paths in the same model (e.g., Hunter et al., 2019; Sonnentag et al., 2018). Yet, by modeling these mechanisms in parallel, there is an assumption that these paths convey the positive and negative effects of intrusions onto subsequent outcomes independently. By advancing theory about how belongingness stemming from intrusions' social aspect can buffer the negative effect of depletion on job satisfaction and finding support for it, we show that these positive and negative mechanisms can actually interact. Thus, we extend theory by showing that work intrusions may have a more complicated effect on outcomes than previously thought and that understanding this effect may require a joint consideration of both the bright and dark consequences of work intrusions. This finding also answers the call for an integration of interaction effects in the study of dynamic well-being (Sonnentag, 2015).

Fourth, the above interaction effect also adds to self-regulation research on factors that offset the negative effects of self-regulatory resource depletion in the workplace (Lanaj et al., 2016; Lilius, 2012). To our knowledge, this is the first study to show that in a workplace setting, experiencing belongingness can somewhat counteract the effects of self-regulatory resource depletion, at the within-person level. This is a key contribution as depletion has been linked to various deleterious effects. We complement studies that have uncovered person-level factors that buffer daily depletion (e.g., Rosen et al., 2016), by showing that even in the absence of these factors, daily belongingness could be an antidote to the feeling of depletion in the workplace.

Practical Implications

Given the widespread occurrence of work intrusions, insight into how employees are affected by intrusions can help managers better manage employees' workplace experience. In the past, scholars and popular writers have often recommended the reduction of work intrusions, based on the negative effects of work intrusions found in prior research (Baethge & Rigotti, 2013; Lin et al., 2013; Perlow, 1999; Schulte, 2015; Steinhilber, 2017). However, our study shows that work intrusions could also have potential beneficial effects, indicating that reducing all work intrusions may, in fact, eliminate some of these potential benefits. Thus, we call for better management, rather than the complete elimination, of these workplace occurrences.

Specifically, work intrusions can have beneficial effects by providing an avenue for the fulfillment of interrupted employees' need to belong. Our supplemental analyses indicated that high task interdependence and high work scheduling autonomy may potentially boost the positive association between work intrusions and belongingness at the day level. It could be that emphasizing these components more may help unlock the beneficial effects of work intrusions' social aspect even more by promoting belongingness, which can positively affect employee job satisfaction and also buffer the negative effects of self-regulatory depletion on job satisfaction.

Our results also showed that self-regulatory resource depletion is an important reason for the negative effects of work intrusions. Since the same central store of self-regulatory resources is used to fuel various acts of self-control (Baumeister & Vohs, 2016), it is

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possible that even interventions that are not targeted specifically at managing work intrusions per say, but which are focused more generally on reducing self-regulatory demands in the work environment (e.g., redesigning workplaces to reduce distracting background noises or implementing less stringent norms of emotional labor in the workplace), could help reduce the negative effects of work intrusions. The reason is that if other general workplace demands on employees' self-regulatory resources are lessened, they will have more resources available to handle the self-regulatory demands of work intrusions, reducing the negative impact of intrusions on their well-being.

In sum, given the inevitability of being interrupted by others in the workplace and the bright side of intrusions unearthed in our study, we call on managers to adopt a more balanced view of work intrusions—one that not only focuses on intrusions' negative effects on employee well-being but also acknowledges intrusions' potential benefits for employee well-being.

Limitations and Future Research

There are two approaches to studying work intrusions—frequency and episodic (Puranik et al., 2020). We use the former to study the aggregate effect of multiple intrusions. This in line with the call by Baethge et al. (2015) who noted that given the large number of intrusions in the workplace, it is crucial to study intrusions' cumulative effects. Plus, the focus on the frequency of work intrusions is aligned with recent work in this area (e.g., Fletcher et al., 2018; Hunter et al., 2019; Lin et al., 2013; Parke et al., 2018; Sonnentag et al., 2018). A frequency approach also makes sense here because our hypothesized mediators—self-regulatory resource depletion and belongingness—tend to be affected by the aggregate effect of multiple episodes employees encounter over a day (e.g., Bernstein et al., 2018; Lanaj et al., 2016; Milek et al., 2018). This approach, however, is less suitable for studying any single intrusion episode in detail, which may be needed when interest is in event-level attributes of intrusions. An episodic approach that considers each intrusion separately (e.g., Beal et al., 2005) and thus allows one to zoom in on the specific contents of each work intrusion such as, say, the attributes of the social interaction with the interrupter (e.g., duration, complexity, valence, topic, etc.) may help in such cases.

Another way of exploring whether different work intrusions have different effects is by checking if the work intrusion construct has different underlying dimensions defined by different types of intrusions. Our exploratory analyses of the two potential factors underlying the Parke et al. (2018) scale may be relevant in this regard. For instance, factor 1, which primarily composed of only task-related intrusions, was related to depletion but not belongingness, while factor 2, which was made of one nonwork intrusion item and one item capturing intrusions for work-related updates, was related to belongingness but not depletion. This seems to hint that intrusions related to work tasks may predominantly result in depletion due to high attention-switching demands, but may not facilitate belongingness; while intrusions that are nonwork-related or which lead to information gain could facilitate belongingness as they might involve a significant social interaction component, but may not result in depletion. In other words, there could potentially be a difference in being interrupted for social reasons as opposed to being interrupted for work-related help or information or for being assigned new tasks. Thus, although scholars have treated work intrusion as a unidimensional construct in the past (e.g., Jett & George, 2003; Lin et al., 2013; Parke et al., 2018; Rogers & Barber, 2019), the aforementioned results of our exploratory analysis hint that focusing on the reason for the intrusion could be a way of delineating different dimensions underlying the work intrusion construct.

However, our exploratory analysis is not without shortcomings. A theoretical concern with this analysis is that we do not have well-defined theoretical constructs underlying each of these factors (e.g., factor 2 is a mix of work and nonwork intrusions and so, its theoretical meaning is not entirely clear). An empirical concern is that splitting the presently unidimensional Parke et al. (2018) scale into separate 3-item and 2-item scales may not be ideal from a psychometric standpoint (Kline, 2016), especially given the low reliability of the 2-item measure (average Cronbach alpha and omega were both .53). Plus, the smaller number of items could also mean that these scales underspecify the content domain for different work intrusion types. These limitations notwithstanding, the exploration of different dimensions of work intrusions and the relationship of these dimensions with key outcomes is a crucial avenue for future research that can provide a fine-grained understanding of work intrusions and their effects.

As one example of this, scholars could examine the work/nonwork distinction as one way to classify the different dimensions of intrusions. Work-related intrusions could perhaps instill a sense of competence, while nonwork intrusions could maybe provide respite from the monotony of daily work. Alternately, scholars could also classify work intrusions based on whether they are in-person or virtual in nature. In-person work intrusions include a rich set of nonverbal cues and may also require immediate response from the interrupted person, whereas work intrusions occurring virtually (e.g., instant messenger chat) may at times lack these factors. There is thus substantial scope to tease apart potential underlying dimensions of work intrusions.

Scholars can also extend our research by exploring boundary conditions for our model. Although we have outlined how our mediating mechanisms interact, there can be other factors that shape the intrusion experience. Daily workload could be one factor, with demands linked to high workload adding to intrusions' negative relationship with depletion and reducing the positive one with belongingness. Employees' occupation could be another factor that shapes the experience of intrusions. For some occupations (e.g., nurses) the stakes of errors are more, and this could shape employee reactions to intrusions. Alternately, one's occupation could affect the level of "control" one has over intrusions, in turn, impacting one's response to intrusions. For instance, we found high task interdependence and work scheduling autonomy seem to bolster the positive effects of intrusions' social aspect. It is also possible that occupations differ in the extent to which people are exposed to similar versus different types of intrusions. Or there could be variation in the time, duration, or weekly/daily distribution of work intrusions across different occupations that could influence the experience of work intrusions. We invite scholars to build further on our findings to explore how work intrusions affect employees in different work contexts and occupations.

Lastly, an interesting point was the lack of interaction effect of belongingness on the relationship between stress and job satisfaction, which, though not formally hypothesized, could be evidence of the conceptual difference between the depletion and stress pathways in our model. Although stress and self-regulatory resource depletion

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are related, they are still distinct constructs with differing physiological and psychological experiences (see Baumeister & Vohs, 2016). In the case of work intrusions specifically, the eliciting conditions for stress (e.g., time pressure; Jett & George, 2003) differ from those responsible for depletion (e.g., self-regulatory demands associated with (re)switching attention between tasks and regulating one's negative reactions to hindered goal progress; Lin et al., 2013; Muraven & Baumeister, 2000). Given this, the difference in the interaction effect of belongingness on the relationship between stress and job satisfaction and that between depletion and job satisfaction seems to suggest that the positive experience of belongingness may perhaps be more relevant in counteracting the effects of self-regulatory resource depletion (e.g., by boosting self-regulatory capacity; Blackhart et al., 2011) than buffering the negative effects of stress. Thus, future researchers have the opportunity to probe if other factors can counteract the negative effects of stress by addressing the eliciting condition of stress (i.e., time pressure) in case of work intrusions. Perhaps people's style of managing their daily schedule or their level of trait polychronicity (i.e., tendency of doing multiple things simultaneously) could be relevant variables to explore in such instances (e.g., Pachler et al., 2018; Parke et al., 2018).

Conclusion

Given their widespread occurrence in the workplace, understanding the consequences of work intrusions has theoretical and practical significance. By expanding the research focus onto work intrusions' social aspect, we provide a comprehensive view of this workplace phenomenon and also contribute to the recent research that has started exploring the positive effects of work intrusions. We do this while not ignoring the task-based aspect of work intrusions and its negative effects uncovered by prior research. The result is a balanced view of work intrusions that sets the stage for more research on the social aspect and the bright side of work intrusions.

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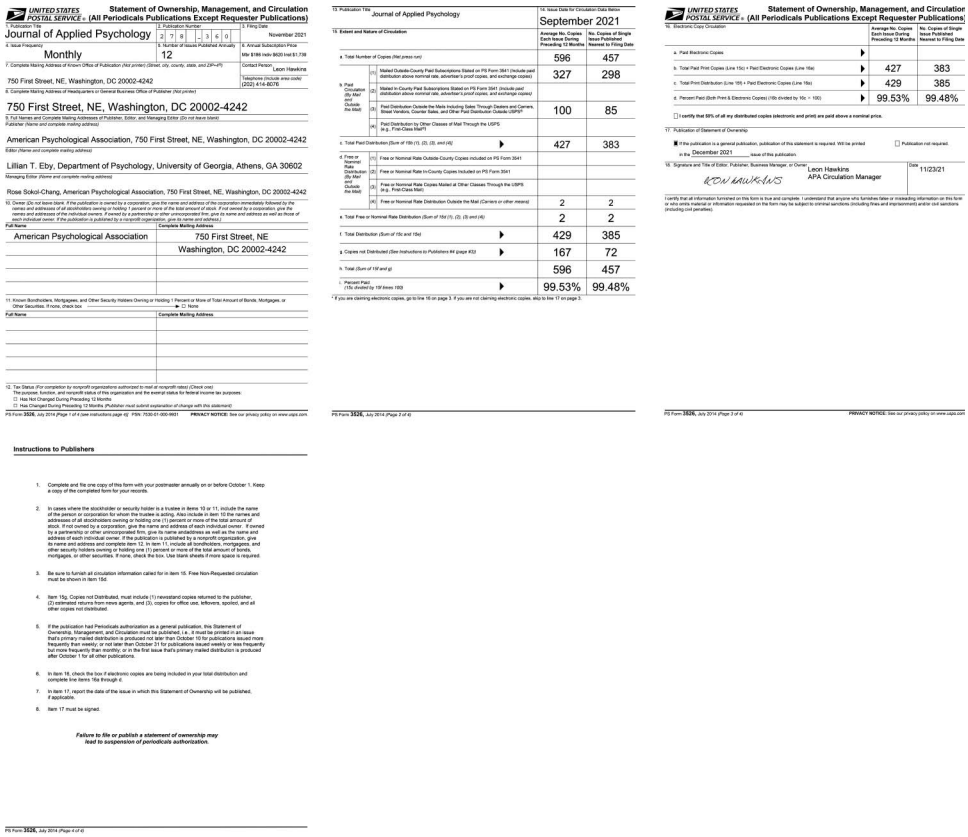

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